
A Median Perspective on Unlabeled Data for Out-of-Distribution Detection

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Abstract

1 Out-of-distribution (OOD) detection is a core requirement for reliable deployment
2 of machine learning systems under distributional uncertainty. Recent work has ex-
3 amined unlabeled auxiliary data as a mechanism for improving OOD detection, but
4 unlabeled in-the-wild data is typically a heterogeneous mixture of in-distribution
5 (InD) and OOD samples. Because this mixture does not provide a clean OOD
6 subset, it is nontrivial to construct an OOD classifier with well-specified negative
7 examples. We introduce Medix, a median-centric framework that identifies candi-
8 date outliers from unlabeled data through a median-based filtering operation.
9 The median is used as a robust estimator of central tendency in gradient space,
10 which makes the filtering criterion less sensitive to noise and extreme observations.
11 The resulting candidate outliers are combined with labeled InD data to train a
12 binary OOD detector. We derive finite-sample error bounds showing that Medix
13 attains low filtering error under explicit concentration and contamination condi-
14 tions. Empirically, Medix improves over existing methods across open-world OOD
15 benchmarks, providing experimental support for the theoretical characterization.

16 1 Introduction

17 When machine learning models are deployed outside controlled training environments, their safety
18 and reliability depend on how they process inputs drawn from out-of-distribution (OOD) sources.
19 Such inputs may correspond to semantic categories that are absent from the training label space and
20 therefore should not receive ordinary in-distribution predictions. Standard neural classifiers, however,
21 often assign confident predictions to OOD inputs and generally do not include an explicit mechanism
22 for rejecting them (Nguyen et al., 2015).

23 Detecting OOD samples at inference time is technically difficult because the unknown test distribu-
24 tions are not observed during supervised training, so the classifier has no direct basis for separating
25 them from in-distribution (InD) inputs. To mitigate this limitation, recent approaches (Katz-Samuels
26 et al., 2022a; Du et al., 2024a) use additional “in-the-wild” data as an auxiliary signal for OOD
27 detection. In particular, Katz-Samuels et al. (2022a) regularize training with unlabeled wild data while
28 preserving the primary objective of classifying labeled InD examples. This data source is practically
29 attractive because it can be collected after deployment from the model’s operating environment, and
30 it can therefore approximate the types of non-training inputs that appear at test time.

31 The central technical obstacle is that unlabeled wild data is not a labeled OOD sample set; it is a
32 mixture containing both InD and OOD observations. Consequently, an OOD detector trained on
33 this source observes samples from a latent mixture distribution without sample-level indicators for
34 membership in the InD or OOD component. This ambiguity makes robust detector construction
35 substantially more constrained than settings with clean auxiliary outliers. The in-the-wild formulation
36 remains comparatively underdeveloped, and formal guarantees are especially limited; to the best of

our knowledge, Du et al. (2024a) is the only prior work that provides a theoretical foundation for this setting.

In parallel, recent data-pruning results indicate that median-based procedures can provide robust selection behavior under contamination (Acharya et al., 2024). Motivated by this observation, we study the following technical question:

How can median-based methods leverage unlabeled wild data to facilitate OOD detection with theoretical guarantees?

We answer this question by formulating OOD detection from unlabeled mixtures through a median-centric optimization perspective. Specifically, we propose Medix, an algorithm that uses median-based gradient filtering to extract candidate OOD samples from wild data with low error. We then analyze the filtering rule and provide one of the few theoretical treatments of the unlabeled “in-the-wild” setting. The analysis bounds the fraction of InD samples incorrectly flagged as outliers and decomposes this error into two terms: a contamination term that quantifies the influence of OOD samples and remains controlled when the OOD proportion is below 50%, and a concentration term that uses sub-Gaussian gradient behavior to control deviations of InD gradients from their mean. Together, these terms imply that, with high probability, the inlier misclassification rate remains small under the stated assumptions.

We evaluate Medix against two method classes: detectors trained only on InD data, and detectors trained with both InD data and an auxiliary unlabeled source. On CIFAR-100 (Krizhevsky et al., 2009), Medix reduces FPR95 by an average of 40.98% relative to the strong KNN+ baseline (Sun et al., 2022). In contrast to Outlier Exposure (Hendrycks et al., 2019), which assumes an auxiliary dataset that is cleanly separable from the InD distribution, Medix operates directly on mixed wild data. Relative to WOODS (Katz-Samuels et al., 2022a), Medix decreases average FPR95 by 1.32% on CIFAR-100 and by 2.60% on CIFAR-10.

Our key contributions are as follows:

- C1)** We propose Medix, a median-centric greedy procedure that filters candidate outliers from unlabeled wild data and subsequently trains an OOD detector on the extracted candidates together with labeled InD samples. The principal technical contribution is the filtering stage.
- C2)** We establish theoretical guarantees for median-based filtering in unlabeled mixtures. In particular, we show that both inlier-to-outlier and outlier-to-inlier misclassification rates are controlled by contamination effects and concentration effects, where the former remain bounded below 50% OOD contamination and the latter quantify finite-sample stability of gradient statistics.
- C3)** We evaluate Medix on eleven InD–OOD pairs against 20 competitive baselines. The results show consistent performance gains across the benchmark suite, and a controlled synthetic study shows that Medix outlier extraction attains a low error rate (e.g., 12.5%; see Figure 2), in agreement with the theoretical analysis.

2 Preliminaries and Problem Setup

This section specifies the OOD detection setting and then defines the data model, classifier parameterizations, loss functions, and evaluation objective.

Labeled In-Distribution Data. Let $\mathcal{X}$ denote the input space and let $\mathcal{Y} = \{1, \dots, K\}$ denote the InD label space. A labeled dataset $\mathcal{S}_{\text{in}} = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)\}$ consists of n i.i.d. samples from $\mathbb{P}_{XY}$, an unknown joint distribution on $\mathcal{X} \times \mathcal{Y}$. The marginal of $\mathbb{P}_{XY}$ on $\mathcal{X}$ is denoted by $\mathbb{P}_{\text{in}}$ and defines the InD input distribution. The dataset $\mathcal{S}_{\text{in}}$ is used to train the InD classifier.

Out-of-distribution detection. We consider a deployment setting in which the model is trained on labeled InD data but subsequently receives inputs from categories outside the training label space, i.e., labels satisfying $y \notin \mathcal{Y}$. For these inputs, the appropriate behavior is rejection rather than assignment to an InD class. Thus, the inference-time decision problem is binary: determine whether an input is sampled from the InD distribution or from an OOD source.

Unlabeled wild data. A primary obstacle in OOD detection is the limited availability of labeled OOD examples. Because the OOD sample space may be extremely broad, exhaustive collection and

86 annotation are generally impractical. Following Katz-Samuels et al. (2022a), we therefore incorporate
 87 unlabeled wild data, $\mathcal{S}_{\text{wild}} = \{\tilde{\mathbf{x}}_1, \dots, \tilde{\mathbf{x}}_m\}$, into the learning framework. This dataset is collected
 88 in the deployment environment of a model pretrained on $\mathcal{S}_{\text{in}}$ and is treated as a mixture of InD and
 89 OOD samples. Following Du et al. (2024a); Katz-Samuels et al. (2022a), we model the marginal
 90 distribution of wild data through the Huber contamination model (Huber, 1964):

$$\mathbb{P}_{\text{wild}} := (1 - \pi)\mathbb{P}_{\text{in}} + \pi\mathbb{P}_{\text{out}}, \quad \pi \in (0, 1], \quad (1)$$

91 where π is the OOD contamination proportion and $\mathbb{P}_{\text{out}}$ is the OOD distribution on $\mathcal{X}$. The degenerate
 92 case $\pi = 0$ contains no OOD samples and therefore reduces to a trivial detection problem.

93 **Models and Loss Functions.** Let $f_\phi : \mathcal{X} \rightarrow \mathbb{R}^K$ denote the InD classifier with parameters $\phi \in \Phi$,
 94 where Φ is its parameter space. The output of f_ϕ is interpreted as a soft class-probability vector over
 95 the $K = |\mathcal{Y}|$ InD classes, and the supervised InD loss is $\ell : \mathbb{R}^K \times \mathcal{Y} \rightarrow \mathbb{R}$. For OOD detection, we
 96 use a separate scalar-valued classifier $g_\theta : \mathcal{X} \rightarrow \mathbb{R}$ with parameters $\theta \in \Theta$. Its binary loss is denoted
 97 by $\ell_b(g_\theta(x), y_b)$, where $y_b \in \mathcal{Y}_b := \{y_+, y_-\}$, with $y_+ > 0$ assigned to InD samples and $y_- < 0$
 98 assigned to OOD samples.

99 **Learning objective.** The learning objective trains the OOD detector g_θ together with the multi-class
 100 classifier f_ϕ using information from both $\mathbb{P}_{\text{in}}$ and $\mathbb{P}_{\text{wild}}$. At test time, performance is reported using
 101 the following metrics:

$$\begin{aligned} \downarrow \text{FPR}(g_\theta) &= \mathbb{E}_{\mathbf{x} \sim \mathbb{P}_{\text{out}}^{\text{test}}} [\mathbb{I}\{g_\theta(\mathbf{x}) = \text{in}\}], \\ \uparrow \text{TPR}(g_\theta) &= \mathbb{E}_{\mathbf{x} \sim \mathbb{P}_{\text{in}}} [\mathbb{I}\{g_\theta(\mathbf{x}) = \text{in}\}], \\ \uparrow \text{Acc}(f_\phi) &= \mathbb{E}_{(\mathbf{x}, y) \sim \mathbb{P}_{\mathcal{X}\mathcal{Y}}} [\mathbb{I}\{f_\phi(\mathbf{x}) = y\}], \end{aligned}$$

102 where $\mathbb{I}\{\cdot\}$ is the indicator function and $\mathbb{P}_{\text{out}}^{\text{test}}$ denotes the OOD test distribution.

103 3 Method: Median-Centric Framework for OOD Detection

104 This section defines **Medix**, a two-stage learning framework for OOD detection with unlabeled wild
 105 data. The framework augments conventional InD-only training by explicitly estimating a candidate
 106 OOD subset from the wild mixture before fitting the detector. It consists of **1) Outlier Extraction**:
 107 a median-based filtering procedure that selects candidate OOD samples from unlabeled wild data
 108 (Section 3.1), and **2) Detector Training**: binary OOD-detector training using labeled InD samples
 109 and the extracted outlier candidates (Section 3.2). For stage 2, we follow the protocol introduced by
 110 Du et al. (2024a). The resulting pipeline separates the latent-mixture problem from the supervised
 111 binary detection problem, yielding a procedure suitable for open-world deployment scenarios in
 112 which explicit OOD labels are unavailable.

113 3.1 Extracting Candidate Outliers from the Wild Data

114 To extract candidate outliers from the wild mixture $\mathcal{S}_{\text{wild}}$, Medix uses an optimization criterion defined
 115 on parameter gradients. These gradients are computed with respect to a classifier f_ϕ trained only on
 116 the labeled InD dataset $\mathcal{S}_{\text{in}}$. The filtering procedure is specified as follows.

117 **Reference gradient estimation from InD data.** The first step estimates an InD reference gradient
 118 from $\mathcal{S}_{\text{in}}$. We train f_ϕ on $\mathcal{S}_{\text{in}}$ by empirical risk minimization (ERM):

$$\phi_{\mathcal{S}_{\text{in}}} \in \arg \min_{\phi \in \Phi} \mathcal{L}_{\mathcal{S}_{\text{in}}}(f_\phi), \quad \text{where } \mathcal{L}_{\mathcal{S}_{\text{in}}}(f_\phi) = \frac{1}{n} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{S}_{\text{in}}} \ell(f_\phi(\mathbf{x}_i), y_i), \quad (2)$$

119 where $\phi_{\mathcal{S}_{\text{in}}}$ denotes the learned parameters. After training, the mean InD gradient $\bar{\nabla}_{\text{in}}$ is computed by
 120 averaging loss gradients with respect to the model parameters over the InD dataset:

$$\bar{\nabla}_{\text{in}} = \frac{1}{n} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{S}_{\text{in}}} \nabla \ell(f_{\phi_{\mathcal{S}_{\text{in}}}(\mathbf{x}_i), y_i). \quad (3)$$

121 This vector $\bar{\nabla}_{\text{in}}$ functions as the reference statistic against which gradient deviations of wild samples
 122 are measured.

Motivation. We posit that adding OOD samples to $\mathcal{S}_{\text{wild}}$ increases the deviation between the wild-set median gradient and the InD mean gradient $\bar{\nabla}_{\text{in}}$. To empirically examine this property, we use CIFAR-10 (Krizhevsky et al., 2009) as the InD dataset and SVHN (Netzer et al., 2011) as the OOD dataset. The initial $\mathcal{S}_{\text{wild}}$ contains 10,000 CIFAR-10 samples that are disjoint from the training set used to learn $\phi_{\mathcal{S}_{\text{in}}}$ and compute $\bar{\nabla}_{\text{in}}$. We then add SVHN samples incrementally and monitor the L_2 deviation between $\bar{\nabla}_{\text{in}}$ and the element-wise median (EWM) of wild-sample gradients:

$$\left\| \bar{\nabla}_{\text{in}} - \text{EWM} \left(\left\{ \nabla \ell \left(f_{\phi_{\mathcal{S}_{\text{in}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i} \right) \right\}_{i \in \mathcal{S}_{\text{wild}}} \right) \right\|.$$

As shown in Figure 1, the measured L_2 deviation increases monotonically as the number of OOD samples grows. This empirical regularity motivates the filtering objective below. It also determines the stopping criterion used by Medix: the iterative procedure terminates when the inter-iteration change in the L_2 deviation falls below the threshold ϵ .

Filtering potential outliers from unlabeled wild data. Motivated by Figure 1, we define the following optimization problem for estimating the InD subset of the wild mixture, and hence the complementary outlier subset $\mathcal{S}_{\text{out}}^*$:

$$\mathcal{S}_{\text{in}}^* = \arg \min_{\mathcal{S} \subseteq \mathcal{S}_{\text{wild}}} \left\| \bar{\nabla}_{\text{in}} - \text{EWM}(G_{\mathcal{S}}) \right\|, \text{ where } G_{\mathcal{S}} = \left\{ \nabla \ell \left(f_{\phi_{\mathcal{S}_{\text{in}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i} \right) \right\}_{i \in \mathcal{S}}. \quad (4)$$

Here $\text{EWM}(\cdot)$ is the element-wise median operator, and $\hat{y}_{\tilde{\mathbf{x}}_i}$ is the pseudo-label predicted for the wild sample $\tilde{\mathbf{x}}_i$. For compact notation, we write $G_{\mathcal{S}} = \{\nabla \ell(\tilde{\mathbf{x}}_i)\}_{i \in \mathcal{S}}$. The objective selects a subset $\mathcal{S}$ whose gradient EWM is closest to the InD reference gradient $\bar{\nabla}_{\text{in}}$. In view of Figure 1, the minimizer $\mathcal{S}_{\text{in}}^*$ is interpreted as the estimated InD component of $\mathcal{S}_{\text{wild}}$, while its complement $\mathcal{S}_{\text{out}}^* = \mathcal{S}_{\text{wild}} \setminus \mathcal{S}_{\text{in}}^*$ is interpreted as the estimated OOD component.

Directly solving (4) is computationally prohibitive for large $\mathcal{S}_{\text{wild}}$. We therefore use a greedy leave-one-out approximation, summarized in Algorithm 1. The algorithm iteratively removes samples whose exclusion most reduces the deviation between the wild-set gradient median and the InD reference gradient.

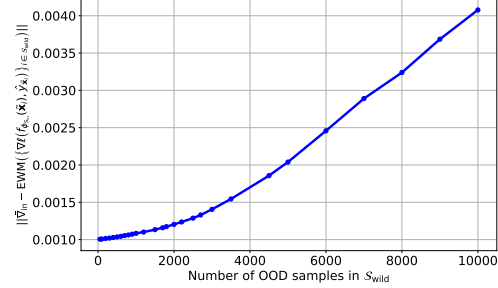


Figure 1: Gradient-median distance deviation as OOD samples are added to $\mathcal{S}_{\text{wild}}$.

At iteration t , the current retained wild set is denoted by $\mathcal{S}$, and the deviation statistic is $d_t = \|\text{EWM}(G_{\mathcal{S}}) - \bar{\nabla}_{\text{in}}\|$. For each candidate sample $i \in \mathcal{S}$, Medix computes the leave-one-out EWM $\text{EWM}(G_{\mathcal{S} \setminus \{i\}})$ and the corresponding distance to $\bar{\nabla}_{\text{in}}$. The decrease induced by removing i is $\delta_i = d_t - \|\text{EWM}(G_{\mathcal{S} \setminus \{i\}}) - \bar{\nabla}_{\text{in}}\|$. The k samples with the largest decreases are assigned to the candidate OOD set and removed from $\mathcal{S}$. Iteration continues until the maximum decrease is no longer significant or the iteration budget is exhausted. The stopping threshold ϵ is based on the change in L_2 distance across iterations: after sufficient OOD removal, the deviation should approach a small value, and further removal would increase the risk of discarding InD samples.

3.2 Training the OOD Detector with Candidate Outliers

After Medix extracts the candidate outlier set $\hat{\mathcal{S}}_{\text{out}}$, we train the OOD detector g_{θ} following the protocol of Du et al. (2024a). The detector objective enforces binary separability around the zero threshold: labeled InD samples $\mathbf{x} \in \mathcal{S}_{\text{in}}$ are encouraged to produce positive outputs, whereas candidate outliers $\tilde{\mathbf{x}} \in \hat{\mathcal{S}}_{\text{out}}$ are encouraged to produce negative outputs. The corresponding objective is

$$\begin{aligned} \mathcal{L}_{\mathcal{S}_{\text{in}}, \hat{\mathcal{S}}_{\text{out}}}(g_{\theta}) &= \mathcal{L}_{\mathcal{S}_{\text{in}}}^+(g_{\theta}) + \mathcal{L}_{\hat{\mathcal{S}}_{\text{out}}}^-(g_{\theta}), \\ \mathcal{L}_{\mathcal{S}_{\text{in}}}^+(g_{\theta}) &= \mathbb{E}_{\mathbf{x} \in \mathcal{S}_{\text{in}}} \mathbb{I}\{g_{\theta}(\mathbf{x}) \leq 0\}, \\ \mathcal{L}_{\hat{\mathcal{S}}_{\text{out}}}^-(g_{\theta}) &= \mathbb{E}_{\tilde{\mathbf{x}} \in \hat{\mathcal{S}}_{\text{out}}} \mathbb{I}\{g_{\theta}(\tilde{\mathbf{x}}) > 0\}. \end{aligned} \quad (5)$$

Because the 0/1 loss is non-differentiable, we optimize a differentiable sigmoid-based binary surrogate that preserves the intended decision-boundary semantics while enabling gradient-based training. The

Algorithm 1 Iterative Outlier Detection via **Medix**

Require: $\bar{\nabla}_{\text{in}}, G_i = \nabla \ell(f_{\phi_{\mathcal{S}_{\text{in}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i})}$, maximum iterations T , hyperparameters ϵ, k

- 1: Initialize $\mathcal{S} \leftarrow \mathcal{S}_{\text{wild}}$ (wild set), $\mathcal{O} \leftarrow \emptyset$ (outliers), $t \leftarrow 0$ (iteration), $d_t \leftarrow 0$, $\delta_{\text{max}} \leftarrow \infty$ (deviations), $\mathcal{I}_k \leftarrow \emptyset$
- 2: **while** $t \leq T$ or $|\delta_{\text{max}}| > \epsilon$ **do**
- 3: $\mathcal{O} \leftarrow \mathcal{O} \cup \{\tilde{\mathbf{x}}_i : i \in \mathcal{I}_k\}$ ▷ Add outliers to set $\mathcal{O}$
- 4: $d_t \leftarrow \|\text{EWM}(G_{\mathcal{S}}) - \bar{\nabla}_{\text{in}}\|$ ▷ Compute L2 deviation
- 5: **for each** $i \in \mathcal{S}$ **do**
- 6: $\delta_i \leftarrow d_t - \|\text{EWM}(G_{\mathcal{S} \setminus \{i\}}) - \bar{\nabla}_{\text{in}}\|$
- 7: **end for**
- 8: $\mathcal{I}_k \leftarrow \text{indices}(\text{top-}k(\{\delta_i\}_{i \in \mathcal{S}}))$ ▷ Select top- k indices
- 9: $\mathcal{S} \leftarrow \mathcal{S} \setminus \mathcal{I}_k$ ▷ Remove outliers from $\mathcal{S}$
- 10: $\delta_{\text{max}} \leftarrow \max_{i \in \mathcal{S}} \{\delta_i\}$, $t \leftarrow t + 1$
- 11: **end while**
- 12: **return** $\mathcal{S}_{\text{out}} = \mathcal{O}$ ▷ Return detected outliers

167 full training objective also includes the InD risk in Equation 2, so detector learning does not discard
 168 the multi-class InD classification objective. This joint optimization yields an OOD detector g_θ that is
 169 trained to reject samples from $\mathbb{P}_{\text{out}}$ while maintaining InD predictive performance.

170 4 Theoretical Analysis

171 We next state theoretical guarantees for the Medix filtering stage. The results provide high-probability
 172 upper bounds on both InD samples incorrectly removed as outliers and OOD samples incorrectly
 173 retained as inliers. Taken together, these bounds characterize the two-sided robustness of EWM-based
 174 filtering. Full proofs are provided in Appendix C.

Theorem 4.1 (Inlier Misclassification Bound). *Assume that the gradients of InD points in $\mathcal{S}_{\text{wild}}$ are i.i.d., and each coordinate is sub-Gaussian with variance proxy σ^2 . Let $\epsilon = \sigma\sqrt{2\log(2dm_{\text{in}})}$, and fix any confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification rate of the EWM filtering rule satisfies:*

$$\text{ERR}_{\text{in}} \leq \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}.$$

175 Theorem 4.1 decomposes the inlier misclassification rate into two terms. The *concentration term*
 176 captures finite-sample fluctuations of InD gradients and decreases with sample size, whereas the
 177 *contamination term* quantifies the displacement of the EWM induced by OOD samples. The bound
 178 remains controlled when the contamination ratio satisfies $\pi < 0.5$, which is the regime in which a
 179 median statistic retains its robust-center interpretation. We now analyze the complementary error
 180 event: true outliers that are not flagged and are therefore retained as inliers.

Theorem 4.2 (Outlier Misclassification Bound). *Assume that the gradients of OOD points in $\mathcal{S}_{\text{wild}}$ are i.i.d., and each coordinate is sub-Gaussian with variance proxy σ_{out}^2 . Suppose the mean OOD gradient μ_{out} satisfies a separation condition:*

$$\|\mu_{\text{out}} - \bar{\nabla}_{\text{in}}\|_2 \geq \Delta\sqrt{d},$$

for some $\Delta > 0$. Then, for any tolerance $\epsilon \in (0, \Delta)$ and confidence level $\delta \in (0, 1)$, with probability at least $1 - \delta$, the outlier misclassification rate satisfies:

$$\text{ERR}_{\text{out}} \leq \underbrace{2d \exp\left(-\frac{(\Delta - \epsilon)^2}{2\sigma_{\text{out}}^2}\right)}_{\text{Separation term}} + \underbrace{\sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}}_{\text{Concentration term}} + \underbrace{\frac{1 - \pi}{2\pi}}_{\text{Contamination term}}.$$

181 Theorem 4.2 bounds the fraction of OOD points that remain in the retained set. In combination with
 182 Theorem 4.1, it yields a two-sided guarantee for Medix filtering. The two error rates are determined
 183 by the interaction of the following effects:

184 **Contamination Effect.** In Theorem 4.1, the median criterion remains stable under a nontrivial OOD
 185 fraction provided that $\pi < 0.5$. The term $\pi/[2(1 - \pi)]$ increases with π and represents the extent to
 186 which OOD gradients can shift the EWM. Conversely, Theorem 4.2 contains the penalty $\frac{1-\pi}{2\pi}$, which
 187 formalizes the difficulty of recovering OOD points when the OOD component is underrepresented.

188 **Concentration Effect.** Both bounds use sub-Gaussian concentration of gradient coordinates. For
 189 InD samples, concentration around $\bar{\nabla}_{\text{in}}$ stabilizes the EWM in finite samples. For OOD samples,
 190 concentration around μ_{out} permits an explicit bound on the probability that an OOD gradient falls
 191 sufficiently close to the InD reference to evade filtering. These finite-sample terms scale as $1/\sqrt{m}$,
 192 where m denotes the relevant component sample size.

193 **Separation Effect.** The separation term appears only in Theorem 4.2 and specifies how far the
 194 OOD mean gradient must be from the InD mean gradient for reliable rejection. The exponential
 195 factor $\exp(-(\Delta - \epsilon)^2/2\sigma_{\text{out}}^2)$ expresses this dependence: larger gradient-mean separation reduces
 196 the probability that OOD gradients satisfy the inlier-like filtering criterion.

Theorems 4.1 and 4.2 jointly imply that median filtering separates InD and OOD components in unlabeled mixtures with high probability. InD removal error is controlled by concentration and contamination terms, whereas OOD retention error is controlled by separation, concentration, and reverse-contamination terms. These results provide a formal error characterization for Medix under the stated assumptions.

197 *Remark 4.3. Sub-Gaussian Assumption for Gradient Coordinates.* The coordinate-wise sub-Gaussian
 198 assumption on InD gradients is used to obtain concentration bounds for empirical gradient statistics.
 199 Figure 4a provides empirical support for this modeling assumption: the InD gradient histogram is
 200 concentrated around its mean and exhibits light tails, which is consistent with sub-Gaussian behavior.
 201 Figure 4b further compares empirical gradient quantiles with Gaussian quantiles through a Q-Q
 202 plot (Wilk and Gnanadesikan, 1968); the approximate alignment with the 45-degree reference line
 203 indicates that the empirical distribution is close to Gaussian in the examined coordinates. Since
 204 Gaussian random variables form a subclass of sub-Gaussian random variables, this evidence is
 205 compatible with the assumption. *We also provide a weaker version of the bounds that removes*
 206 *the sub-Gaussian assumption and requires only bounded second moments. This result appears as*
 207 *Theorem C.3 (Appendix C.3). Although the rates deteriorate, the qualitative robustness guarantee for*
 208 *Medix remains valid in this broader setting.*

210 5 Experiments

211 This section evaluates Medix across multiple InD–OOD dataset pairs and compares it with 20
 212 established baselines. All experiments are conducted on NVIDIA A100-SXM4-80GB GPUs. Code
 213 required to reproduce the results will be released upon acceptance.

214 5.1 Models, Datasets, and Baselines

215 **Datasets.** We follow the experimental protocol of Katz-Samuels et al. (2022a), which introduced
 216 OOD-detector learning with wild data, to ensure comparability with prior work. CIFAR-10 and
 217 CIFAR-100 serve as InD datasets ($\mathbb{P}_{\text{in}}$). For OOD evaluation, we use natural image datasets including
 218 PLACES365 (Zhou et al., 2017), SVHN, TEXTURES (Cimpoi et al., 2014), and LSUN-RESIZE &
 219 LSUN-C (Yu et al., 2015) as OOD distributions ($\mathbb{P}_{\text{out}}$). Wild data ($\mathbb{P}_{\text{wild}}$) is simulated by mixing a
 220 subset of InD data ($\mathbb{P}_{\text{in}}$) with OOD data ($\mathbb{P}_{\text{out}}$) using the default contamination parameter $\pi = 0.5$.
 221 For instance, when PLACES365 is the OOD test set, the corresponding wild mixture is constructed
 222 by combining CIFAR samples with PLACES365 samples, and PLACES365 is also used for OOD
 223 testing. The same construction is applied across all OOD datasets and baselines. Each CIFAR InD
 224 dataset is partitioned into two halves: the first 25,000 images train $\phi_{S_{\text{in}}}$, and the remaining images
 225 contribute to the wild mixture S_{wild} . Gradients are computed with respect to the penultimate-layer
 226 weights, which have been shown to provide informative OOD-detection signals (Huang et al., 2021a).

Evaluation metrics. We report three standard metrics: (1) False Positive Rate (FPR95↓) for OOD samples at 95% True Positive Rate on InD samples, (2) Area Under the Receiver Operating Characteristic Curve (AUROC↑), and (3) InD Classification Accuracy (InD Acc↑).

Baselines. Baselines are grouped according to whether they use only InD data or both InD and wild data during training. For InD-only methods ($\mathbb{P}_{in}$), we compare against Maximum Softmax Probability (MSP) (Hendrycks and Gimpel, 2016), ODIN (Liang et al., 2018), Mahalanobis Distance (Lee et al., 2018a), Energy Score (Liu et al., 2020b), ReAct (Sun et al., 2021), DICE (Sun and Li, 2022), KNN Distance (Sun et al., 2022), and ASH (Djurisic et al., 2022), all evaluated using a model trained with softmax cross-entropy. We also include contrastive-loss methods, CSI (Tack et al., 2020a) and KNN+ (Sun et al., 2022), to broaden the comparison. For methods that use both InD and wild data, we compare with Outlier Exposure (OE) (Hendrycks et al., 2019) and energy-regularized learning (Liu et al., 2020b), which impose lower confidence or higher energy on outlier data. We also compare with WOODS (Katz-Samuels et al., 2022a), which introduced wild unlabeled data for OOD detection through constrained optimization. Recent baselines, including CONJ (Peng et al., 2024) and DRL (Zhang et al., 2024), are included to provide a more complete empirical reference.

5.2 Experimental Setup

Following WOODS (Katz-Samuels et al., 2022a), the InD classifier $\phi_{S_{in}}$ is a Wide ResNet (Zagoruyko, 2016) with 40 layers and width factor 2. It is trained by stochastic gradient descent with momentum 0.9, weight decay 0.0005, and initial learning rate 0.1 for 100 epochs, using cosine learning-rate decay, batch size 128, and dropout rate 0.3. The Medix hyperparameters ϵ and k are selected from $\{5e-5, 5e-4, 5e-3, 5e-2\}$ and $\{4k, 7k, 10k, 20k\}$, respectively, with dataset size and OOD performance used as selection criteria. The OOD classifier g_θ is initialized from the pretrained InD classifier $\phi_{S_{in}}$, and a linear binary-classification head is attached to the penultimate-layer representation. This classifier is fine-tuned for 100 epochs with learning rate 0.001 according to Equation (5). The binary classification loss is combined with the InD classification loss, with weight 10 assigned to the binary component. All remaining training parameters match those used for $\phi_{S_{in}}$.

Table 1: OOD detection performance of Medix and baselines with CIFAR-10 as InD data. Metrics are averaged over five runs, and best values are highlighted in **bold**.

Methods	OOD Datasets												InD ACC↑
	SVHN		PLACES365		LSUN-C		LSUN-RESIZE		TEXTURES		Average		
	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	
Using $\mathbb{P}_{in}$ only													
MSP	48.49	91.89	59.48	88.20	30.80	95.65	52.15	91.37	59.28	88.50	50.04	91.12	94.84
ODIN	33.35	91.96	57.40	84.49	15.52	97.04	26.62	94.57	49.12	84.97	36.40	90.61	94.84
Mahalanobis	12.89	97.62	68.57	84.61	39.22	94.15	42.62	93.23	15.00	97.33	35.66	93.34	94.84
Energy	35.59	90.96	40.14	89.89	8.26	98.35	27.58	94.24	52.79	85.22	32.87	91.73	94.84
KNN	24.53	95.96	25.29	95.69	25.55	95.26	27.57	94.71	50.90	89.14	30.77	94.15	94.84
ReAct	40.76	89.57	41.44	90.44	14.38	97.21	33.63	93.58	53.63	86.59	36.77	91.48	94.84
DICE	35.44	89.65	46.83	86.69	6.32	98.68	28.93	93.56	53.62	82.20	34.23	90.16	94.84
ASH	6.51	98.65	48.45	88.34	0.90	99.73	4.96	98.92	24.34	95.09	17.03	96.15	94.84
CSI	17.30	97.40	34.95	93.64	1.95	99.55	12.15	98.01	20.45	95.93	17.36	96.91	94.17
KNN+	2.99	99.41	24.69	94.84	2.95	99.39	11.22	97.98	9.65	98.37	10.30	97.99	93.19
Using $\mathbb{P}_{in}$ and $\mathbb{P}_{wild}$													
OE	1.13	99.53	19.48	94.88	1.91	98.16	0.54	98.84	7.75	98.56	6.16	97.99	94.12
Energy (w/ OE)	5.24	98.72	14.66	96.18	2.35	99.30	4.85	98.62	10.51	97.10	7.52	97.98	94.24
WOODS	0.17	99.91	10.19	98.05	0.31	99.14	0.11	99.38	6.21	98.13	3.40	98.92	94.74
Medix	0.06	99.98	2.98	99.10	0.01	99.98	0.01	99.98	0.96	99.66	0.80	99.74	93.58
(Ours)	± 0.01	± 0.01	± 0.29	± 0.09	± 0.01	± 0.00	± 0.01	± 0.01	± 0.13	± 0.06	± 0.09	± 0.03	± 0.64

5.3 Results

Table 2 reports the CIFAR-100 results, where Medix outperforms all evaluated OOD detection baselines. Two observations are central. First, methods trained with both InD and wild data generally outperform methods trained only on InD data, indicating that wild data provides useful OOD-related information. Relative to KNN+, Medix reduces FPR95 by 52.31% on PLACES365 and 38.24% on TEXTURES. Second, Medix also improves over methods that already use $\mathbb{P}_{wild}$: on CIFAR-100, it obtains an average FPR95 of 5.42%, corresponding to a 1.32% improvement over WOODS, while maintaining InD accuracy of 73.33%. The accuracy gap relative to some InD-only baselines is attributable to the fact that Medix trains on 25,000 labeled InD samples, whereas those baselines use the full 50,000-image CIFAR-100 training set. Table 1 gives the CIFAR-10 results. Medix again exceeds all baselines, reducing FPR95 relative to WOODS by 7.21% on PLACES365 and 5.25% on TEXTURES.

Table 2: OOD detection performance of Medix and baselines with CIFAR-100 as InD data. Metrics are averaged over five runs, and best values are highlighted in **bold**.

Methods	OOD Datasets												InD ACC $\uparrow$
	SVHN		PLACES365		LSUN-C		LSUN-RESIZE		TEXTURES		Average		
	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	
Using $\mathbb{P}_{in}$ only													
MSP	84.59	71.44	82.84	73.78	66.54	83.79	82.42	75.38	83.29	73.34	79.94	75.55	75.96
ODIN	84.66	67.26	87.88	71.63	55.55	87.73	71.96	81.82	79.27	73.45	75.86	76.38	75.96
Mahalanobis	57.52	86.01	88.83	67.87	91.18	69.69	21.23	96.00	39.39	90.57	59.63	82.03	75.96
Energy	85.82	73.99	80.56	75.44	35.32	93.53	79.47	79.23	79.41	76.28	72.12	79.69	75.96
KNN	66.38	83.76	79.17	71.91	70.96	83.71	77.83	78.85	88.00	67.19	76.47	77.08	75.96
ReAct	74.33	88.04	81.33	74.32	39.30	91.19	79.86	73.69	67.38	82.80	68.44	82.01	75.96
DICE	88.35	72.58	81.61	75.07	26.77	94.74	80.21	78.50	76.29	76.07	70.65	79.39	75.96
ASH	21.36	94.28	68.37	71.22	15.27	95.65	68.18	85.42	40.87	92.29	42.81	87.77	75.96
CSI	64.70	84.97	82.25	73.63	38.10	92.52	91.55	63.42	74.70	92.66	70.26	81.44	69.90
KNN+	32.21	93.74	68.30	75.31	40.37	86.13	44.86	88.88	46.26	87.40	46.40	86.29	73.78
Using $\mathbb{P}_{in}$ and $\mathbb{P}_{wild}$													
OE	2.86	99.05	40.21	88.75	4.13	99.05	1.25	99.38	22.86	94.63	14.26	96.17	73.38
Energy (w/ OE)	2.71	99.34	34.82	90.05	3.27	99.18	2.54	99.23	30.16	94.76	14.70	96.51	72.76
WOODS	0.17	99.80	21.87	93.73	0.48	99.61	1.24	99.54	9.95	95.97	6.74	97.73	73.91
Medix	0.16	99.96	15.99	95.23	0.13	99.98	0.83	99.83	8.02	97.79	5.42	98.96	73.33
(Ours)	± 0.02	± 0.00	± 0.66	± 0.14	± 0.06	± 0.02	± 0.36	± 0.06	± 0.75	± 0.30	± 0.37	± 0.10	± 0.83

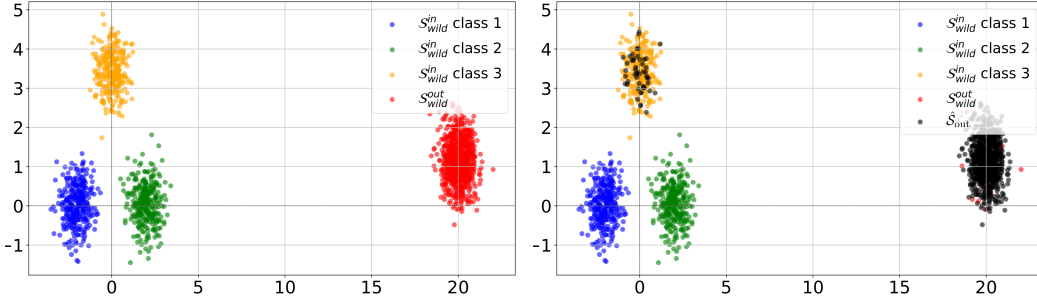


Figure 2: Representative application of Medix to unlabeled wild data. (a) Construction of the wild mixture with InD data $\mathcal{S}_{wild}^{in}$ and OOD data $\mathcal{S}_{wild}^{out}$, where inliers are sampled from three multivariate Gaussian distributions. (b) Outliers $\hat{\mathcal{S}}_{out}$ selected by Medix (black), for which the fraction of selected samples belonging to $\mathcal{S}_{wild}^{in}$ is 12.5%.

A representative visual example of Medix. We further examine the outlier-extraction stage in Algorithm 1 using a two-dimensional synthetic mixture that permits direct visualization. InD data are sampled from three multivariate Gaussian distributions corresponding to three classes, with mean vectors $[-2, 0]$, $[2, 0]$, and $[0, 2\sqrt{3}]$. All three classes use covariance matrix $0.25 \cdot I$. OOD data are sampled from $\mathcal{N}([20, 2\sqrt{3}], 0.25 \cdot I)$. Figure 2 shows that the selected outliers largely match the ground-truth OOD component. Medix flags 87.5% of actual OOD samples as outliers, demonstrating robust extraction behavior in this controlled setting.

Additional studies and insights. Additional experiments are provided in Appendix A. These include: (1) hyperparameter sensitivity analysis (Appendix A.2), showing that Medix is stable across several choices of ϵ and k ; (2) a comparison between EWM and geometric median (Appendix A.1), showing that EWM is more sensitive to distributional shifts and therefore more effective for the proposed filtering criterion; (3) comparisons with methods based on competitive contrastive learning objectives (Appendix A.3); (4) evaluation under a large-scale unseen-OOD setting with $\mathbb{P}_{out}^{test} \neq \mathbb{P}_{out}$ (Appendix A.4); (5) computation and memory profiling for Medix filtering (Appendix A.6); (6) evaluation of pseudo-label quality (Appendix A.5), showing robustness to noisy or low-confidence pseudo-labels; and (7) comparison with semi-supervised open-set recognition methods (Appendix A.7).

6 Related Work

OOD detection has been studied extensively in recent years (Fort et al., 2021; Yang et al., 2024; Fang et al., 2022; Zhu et al., 2022; Yang et al., 2022; Wang et al., 2022c; Galil et al., 2023; Djurisić et al., 2023; Tao et al., 2023; Zheng et al., 2023; Wang et al., 2022b, 2023b; Uppaal et al., 2023; Zhu et al., 2023; Bai et al., 2023; Ming and Li, 2024; Zhang et al., 2023; Ghosal et al., 2024; Abbas et al., 2025). One major class of methods defines OOD scores from model outputs or representations, including distance-based methods (Lee et al., 2018b; Tack et al., 2020b; Ren et al., 2021; Schwag

et al., 2021; Sun et al., 2022; Du et al., 2022a; Ming et al., 2023; Ren et al., 2022), gradient-based scores (Huang et al., 2021b), energy-based scores (Liu et al., 2020c; Wang et al., 2021; Wu et al., 2023), confidence-based scores (Bendale and Boulton, 2016; Hendrycks and Gimpel, 2017; Liang et al., 2018), and Bayesian uncertainty estimates (Gal and Ghahramani, 2016; Lakshminarayanan et al., 2017; Maddox et al., 2019; Malinin and Gales, 2019; Wen et al., 2020; Kristiadi et al., 2020).

Another class modifies training through OOD-oriented regularization (Malinin and Gales, 2018; Geifman and El-Yaniv, 2019; Hein et al., 2019; Meinke and Hein, 2020; Jeong and Kim, 2020; Liu et al., 2020a; Van Amersfoort et al., 2020; Yang et al., 2021; Wei et al., 2022; Du et al., 2022b, 2023; Wang et al., 2023a). Such methods typically penalize overconfidence (Lee et al., 2017; Hendrycks et al., 2019) or increase energy (Liu et al., 2020c; Du et al., 2022c; Ming et al., 2022) on OOD data. Most approaches in this category require an auxiliary OOD dataset.

Several studies (Zhou et al., 2021; Katz-Samuels et al., 2022b; He et al., 2023) relax the labeled-data-only assumption by incorporating unlabeled wild data (Katz-Samuels et al., 2022a; Geng et al., 2025), but they do not specify an explicit outlier-extraction mechanism. By contrast, Du et al. (2024a,b) introduce explicit filtering procedures, although their thresholding strategy differs from the median-centric criterion developed here. In addition, Katz-Samuels et al. (2022a); Du et al. (2024a) assume batch-level mixing, where each batch contains a prescribed InD/OOD ratio. Large outsourced datasets typically do not satisfy this structure, since samples are mixed at the dataset level rather than arranged into controlled batches. Medix addresses this setting by operating directly on dataset-level mixtures. Related positive-unlabeled learning methods train classifiers from positive and/or unlabeled data (Letouzey et al., 2000; Hsieh et al., 2015; Plessis et al., 2015; Niu et al., 2016; Gong et al., 2018; Chapel et al., 2020; Garg et al., 2021; Xu and Denil, 2021; Garg et al., 2022; Zhao et al., 2022; Acharya et al., 2022). The distinction is that these methods primarily separate $\mathbb{P}_{\text{out}}$ from $\mathbb{P}_{\text{in}}$, whereas Medix both identifies outliers in unlabeled data and trains a dedicated OOD detector with provably low filtering error.

7 Conclusions

We introduced Medix, a median-centric framework for OOD detection from unlabeled in-the-wild data. Medix uses the robustness of the median operation to filter candidate outliers from mixed unlabeled data and then trains a binary OOD detector from the extracted candidates and labeled InD samples. The theoretical analysis gives provable bounds on inlier misclassification under OOD contamination up to 50%, with error terms determined by sub-Gaussian concentration and contamination. It also provides complementary bounds on OOD misclassification under explicit separation conditions. Across diverse benchmarks, Medix outperforms 20 baselines: it reduces average FPR95 by 40.98% relative to strong baselines such as KNN+ and improves over state-of-the-art methods including WOODS and DRL, while attaining an outlier-extraction error rate as low as 12.5%.

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584 A Additional Studies

585 A.1 Element-wise Median vs Geometric Median

586 A relevant design choice is the use of the element-wise median rather than
 587 the geometric median (Acharya et al., 2024) as the filtering statistic in Medix.
 588 We evaluate this design choice in a preliminary experiment with CIFAR-10 as the InD dataset and SVHN as the
 589 OOD dataset. The wild set $\mathcal{S}_{\text{wild}}$ contains 5k CIFAR-10
 590 samples disjoint from the data used to train $\phi_{\mathcal{S}_{\text{in}}}$, which is
 591 used to compute $\bar{\nabla}_{\text{in}}$. SVHN OOD samples are then added
 592 incrementally, while we track both the L_2 -norm deviation
 593 between $\bar{\nabla}_{\text{in}}$ and the wild-set gradient median and the frac-
 594 tion of OOD samples removed. Figure 3 shows that, for
 595 the same OOD sample count in the wild set, the element-
 596 wise median identifies a substantially larger fraction of
 597 OOD samples than the geometric median. This behavior
 598 indicates greater sensitivity to the relevant distributional
 599 shift, making EWM the more effective statistic for the
 600 proposed filtering rule.
 601

Table 3: Effect of Medix hyperparameters ϵ and k on OOD detection performance.

Method	FPR95↓	ϵ	k
DICE	88.35	—	—
ASH	21.36	—	—
CSI	64.70	—	—
KNN+	32.21	—	—
OE	2.86	—	—
Energy	2.71	—	—
Medix	0.16	0.005	20000
Medix	0.20	0.0005	20000
Medix	0.68	0.005	10000

602 A.2 Hyperparameter Selection and Sensitivity Analysis

604 We assess Medix sensitivity to the filtering hyperparameters ϵ and k using CIFAR-100 as the InD
 605 dataset and SVHN as the OOD dataset. Table 3 shows that performance is stable over several
 606 hyperparameter settings. With $\epsilon = 0.005$ and $k = 20000$, Medix obtains FPR95 of 0.16. When ϵ is
 607 reduced to 0.0005 or k is reduced to 10000, FPR95 remains competitive at 0.20 and 0.68, respectively.
 608 Thus, the performance degradation is gradual rather than abrupt. These results indicate that although
 609 hyperparameter selection affects the exact operating point, Medix remains effective across a range
 610 of values and continues to outperform the listed baselines. This stability is practically useful when
 611 exhaustive tuning is infeasible.

612 A.3 Comprehensive comparison with recent competitive methods

613 We extend the comparison to competitive baselines with contrastive learning objectives, including
 614 SSD+ (Sehwag et al., 2021), ProxyAnchor (Kim et al., 2020), and CIDER Ming et al. (2023). These
 615 objectives differ from the Medix training formulation, but they provide relevant strong baselines.
 616 We also include recent methods CONJ (Peng et al., 2024), DRL (Zhang et al., 2024), Vim (Wang
 617 et al., 2022a), and VOS (Du et al., 2022c). For this experiment, CIFAR-100 is the InD dataset, and
 618 the ResNet-34 model¹ is trained following Ming et al. (2023); Zhang et al. (2024). Training uses
 619 stochastic gradient descent with momentum 0.9 and weight decay 10^{-4} for 500 epochs. The initial

¹Since ResNet-34 is the standard model across these works, we adopt the same model to ensure consistency and facilitate a fair comparison.

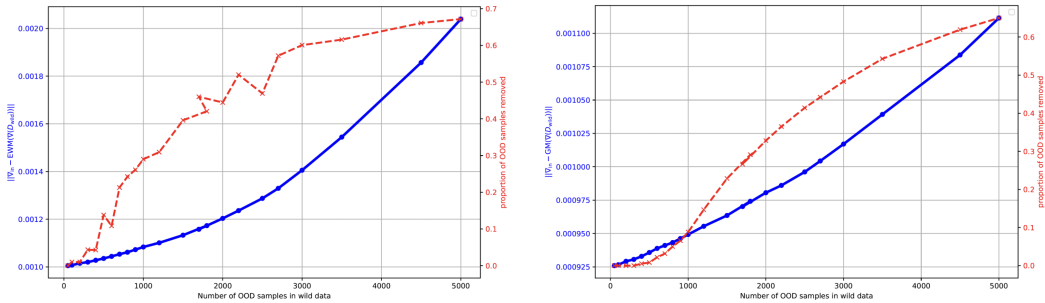


Figure 3: Empirical comparison between element-wise median (EWM) and geometric median (GM) filtering statistics.

Table 4: OOD detection performance of Medix and recent competitive baselines with CIFAR-100 as InD data. Metrics are averaged over five runs, and best values are highlighted in **bold**.

Methods	OOD Datasets									
	SVHN		PLACES365		LSUN-C		TEXTURES		AVERAGE	
	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
MSP	78.89	79.80	84.38	74.21	83.47	75.28	86.51	72.53	83.31	75.46
Mahalanobis	87.09	80.62	84.63	73.89	84.15	79.43	61.72	84.87	79.40	79.70
ODIN	70.16	84.88	82.16	75.19	76.36	80.10	85.28	75.23	78.49	78.85
Energy	66.91	85.25	81.41	76.37	59.77	86.69	79.01	79.96	71.78	82.07
ReAct	50.93	88.75	83.55	73.10	64.02	80.31	64.40	81.95	65.72	81.03
KNN	46.25	90.39	82.08	75.44	60.85	85.61	62.39	83.95	62.89	83.85
Vim	73.42	84.62	85.34	69.34	86.96	69.74	74.56	76.23	80.07	74.98
VOS	43.24	82.80	76.85	78.63	73.61	84.69	57.57	87.31	62.82	83.36
CSI	44.53	92.65	79.08	76.27	75.58	83.78	61.61	86.47	65.20	84.79
ProxyAnchor	87.21	82.43	70.10	79.84	37.19	91.68	65.64	84.99	65.04	84.74
SSD+	31.19	94.19	77.74	79.90	79.39	85.18	66.63	86.18	63.74	86.36
KNN+	39.23	92.78	80.74	77.58	48.99	89.30	57.15	88.35	56.53	87.00
ASH	52.96	90.19	72.62	76.38	75.18	76.52	56.17	86.75	64.23	82.46
CIDER	23.09	95.16	79.63	73.43	16.16	96.33	43.87	90.42	40.69	88.84
CONJ	46.19	90.44	80.81	75.83	60.45	85.90	62.13	83.77	62.40	83.99
DRL	20.15	94.07	76.64	77.55	16.97	94.63	31.97	92.09	36.43	89.59
Medix	0.48	99.87	24.52	93.42	0.73	99.84	8.99	97.92	8.68	97.76
(Ours)	± 0.05	± 0.02	± 1.76	± 0.36	± 0.08	± 0.03	± 0.72	± 0.24	± 0.65	± 0.16

learning rate is 0.5 with cosine scheduling, and the batch size is 512. We use the checkpoints provided by Ming et al. (2023)². Table 4 shows that Medix outperforms these competitive baselines, including contrastive-learning methods, with average FPR95 of 8.68% and average AUROC of 97.76%. In addition, Medix supplies explicit filtering-error guarantees that most alternative approaches do not provide.

A.4 Evaluation on Large-Scale, Complex Unseen OOD Settings

We next evaluate Medix under a large-scale unseen-OOD setting. This setting is technically more demanding for two reasons: (1) it uses a large-scale wild OOD dataset ($\mathbb{P}_{\text{out}}$), which increases data complexity and computational cost; and (2) the test-time OOD distribution ($\mathbb{P}_{\text{out}}^{\text{test}}$) is intentionally different from the wild OOD distribution used during training, i.e., $\mathbb{P}_{\text{out}}^{\text{test}} \neq \mathbb{P}_{\text{out}}$. This protocol more closely approximates deployment settings in which future OOD samples need not match the observed wild outliers. We use CIFAR-100 as labeled InD data, the 300K Random Images dataset Hendrycks et al. (2019) as unlabeled wild OOD data, and SVHN as the test OOD dataset. The protocol is challenging because the wild OOD source (300K Random Images) differs substantially from the test OOD source (SVHN). We compare Medix against baselines that also use wild data during training. Table 5 shows that Medix achieves lower FPR95 (41.29 ± 1.2) and higher AUROC (87.25 ± 0.6) than the baselines, indicating improved separation between InD and unseen OOD data.

Table 5: OOD detection performance under the large-scale unseen-OOD evaluation protocol.

Method	FPR ↓	AUROC ↑
OE	68.80 ± 2.8	82.89 ± 1.1
Energy (w/ OE)	69.81 ± 2.4	85.59 ± 1.0
WOODS	69.41 ± 2.7	86.76 ± 0.8
Medix (ours)	41.29 ± 1.2	87.25 ± 0.9

A.5 Evaluating the Impact of Pseudo-Label Quality on OOD Filtering and Robustness

We examine whether pseudo-label quality, represented by the softmax confidence of $\hat{y}_{\tilde{x}_i}$, affects Medix filtering and whether removing low-confidence pseudo-labels improves robustness. The experiment uses CIFAR-10 as the InD dataset and LSUN-Resize as the OOD dataset. We compute softmax probabilities for pseudo-labeled samples and discard predictions below a confidence threshold of 0.6, which removes 15.98% of the samples from the training set.

Filtering low-confidence pseudo-labels produces essentially unchanged OOD performance. Both the filtered and unfiltered variants obtain FPR95 of 0.01%, while AUROC changes only from 99.98% to 99.99% under filtering. Thus, low-confidence pseudo-label removal does not materially affect

²<https://github.com>

robustness or OOD detection accuracy, indicating that Medix is already resilient to noisy or low-confidence pseudo-labels in this setting.

A.6 Evaluating Computation and Memory Efficiency in Medix Filtering

We profile the computational and memory requirements of the Medix filtering phase on larger unlabeled mixtures. Experiments are conducted on an NVIDIA A100-SXM4-80GB GPU with approximately 15,000 unlabeled samples per InD–OOD pair. CIFAR-10 and CIFAR-100 are used as InD datasets, and LSUN-Resize is used as the OOD dataset. Table 6 shows that peak GPU memory remains modest and that filtering completes within a tractable wall-clock time despite thousands of unlabeled samples. These measurements indicate that Medix filtering is computationally feasible and does not require excessive GPU memory in the evaluated settings.

Table 6: Runtime and GPU-memory profile for Medix filtering on an NVIDIA A100 80GB GPU.

InD–OOD	Wall Clock Time (s)	Peak GPU Memory (MB)	Current GPU Memory (MB)
CIFAR10 – LSUN-Resize	4497.17	99.46	31.74
CIFAR100 – LSUN-Resize	5478.36	99.56	31.79

A.7 Comparison with Semi-Supervised Open-Set Recognition Methods

Medix differs from recent semi-supervised open-set recognition methods (Saito et al., 2021; Fan et al., 2023; Hang and Zhang, 2024; Wang et al., 2024) in objective, data role, and mechanism. In OpenML Saito et al. (2021), the primary objective is to train a standard semi-supervised classifier in the presence of outliers. In our setting, the primary objective is to identify OOD samples in unlabeled wild data and train a dedicated OOD detector. Thus, SSL methods (Saito et al., 2021; Fan et al., 2023; Hang and Zhang, 2024; Wang et al., 2024) aim to improve classification despite outlier contamination, whereas Medix targets open-world OOD detection without labeled OOD data.

The treatment of outliers is also different. SSL methods typically suppress OOD samples as nuisance noise so that the classifier remains robust, while Medix explicitly extracts candidate OOD samples and uses them as negative data for detector training. Consequently, OOD samples are not merely down-weighted; they become part of the detector-construction pipeline.

The algorithmic mechanisms are distinct as well. OpenMatch uses consistency regularization, enforcing stable classifier outputs across stochastic transformations through soft constraints that encourage smooth decision boundaries. Medix instead uses median-based gradient filtering, relying on the robustness of the median to extreme observations. Its theoretical analysis bounds filtering error through contamination terms and concentration terms based on sub-Gaussian InD gradient behavior.

B Broader Impacts and Limitations

OOD detection is an important component of reliable machine learning deployment because deployed systems may encounter inputs outside the training distribution. Medix contributes a median-centric mechanism for improving OOD handling from unlabeled wild data and may be relevant to safety-sensitive domains such as healthcare, autonomous systems, and finance, where erroneous confident predictions can be costly. By improving rejection of distributionally anomalous inputs, the framework addresses one technical barrier to dependable model deployment. Future work could integrate Medix with generative models for outlier synthesis or adapt the filtering criterion to dynamic environments in which OOD distributions evolve over time.

A limitation of Medix is its reliance on an unlabeled mixture that contains both InD and OOD data. In practice, such data may be noisy, corrupted, or sampled inconsistently across deployment environments. This assumption is not specific to Medix; it reflects the uncertainty inherent in wild-data collection and is standard in robust-learning formulations. We make the assumptions explicit in the main theorems and specify the geometric separation condition needed for reliable filtering in Remark 4.3. We also provide a relaxed guarantee in Theorem C.3, which removes the sub-Gaussian tail assumption and requires only weaker moment conditions. Thus, although the analysis uses

696 simplifying assumptions, they are interpretable and partially relaxable, leaving a clear path for future
697 work on OOD detection under more realistic noise models.

698 C Proof of Theorem

699 For consistency across the theoretical derivations and empirical descriptions, Table 7 summarizes the
700 principal notation used in the paper. The table defines gradient quantities, sample counts, filtering-
701 related sets, and parameters appearing in the concentration bounds. Unless otherwise specified, bold
702 symbols denote vectors, $\mathcal{X}$ denotes the input space, and all gradients are computed with respect to a
703 fixed pretrained model f_ϕ .

Table 7: Summary of notation used in the theoretical analysis. Bold symbols denote vectors, and $\mathcal{X}$ denotes the input space.

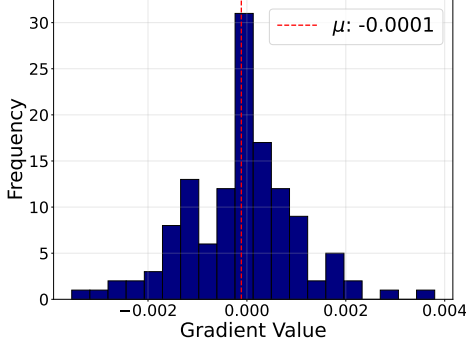
Symbol	Meaning
$\mathcal{X}$	Input space (e.g., images in $[0, 1]^d$)
f_ϕ	Model parameterized by ϕ
$\ell(f_\phi(x))$	Loss function evaluated at input x
$\nabla \ell(f_\phi(x))$	Gradient of loss at point x
$\mathbf{g}(x)$	Shorthand for $\nabla \ell(f_\phi(x))$, gradient vector
$\bar{\mathbf{g}}_{\text{in}}$	Empirical mean of gradients over InD points
m	Total number of unlabeled samples in $\mathcal{S}_{\text{wild}}$
$m_{\text{in}} = (1 - \pi)m$	Number of InD samples
$m_{\text{out}} = \pi m$	Number of OOD samples
$\pi \in (0, 1)$	OOD contamination ratio
$\mathcal{S}_{\text{wild}}$	Unlabeled mixture of InD and OOD data
$\mathcal{S}_{\text{in}}^*$	Inliers selected by the median filter
$\mathcal{S}_{\text{out}}^*$	OOD points selected by the median filter
$\mathcal{B}_\varepsilon(\bar{\mathbf{g}}_{\text{in}})$	Ball of radius ε centered at $\bar{\mathbf{g}}_{\text{in}}$
m_ε	Number of InD points inside $\mathcal{B}_\varepsilon$
m_ε^*	Number of selected inliers inside $\mathcal{B}_\varepsilon$
m_{sep}^*	Number of well-separated OODs in selected set
ERR_{in}	Fraction of InD points misclassified as OOD
ERR_{out}	Fraction of OOD points misclassified as InD
σ^2	Sub-Gaussian proxy variance of each gradient coordinate
ε	Threshold for gradient deviation, $\varepsilon = \sigma \sqrt{2 \log(2dm_{\text{in}})}$
$\delta \in (0, 1)$	Confidence parameter for concentration bounds
d	Dimensionality of gradient vectors

704 C.1 Bound on the Inlier Misclassification Rate

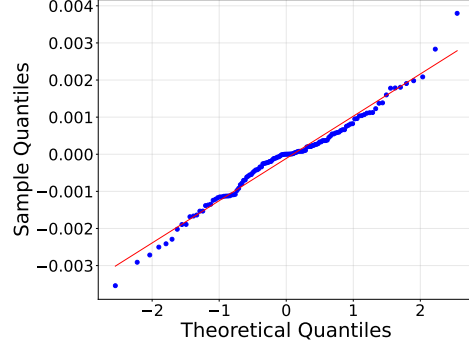
705 **Theorem C.1** (Inlier Misclassification Bound). *Assume that the gradients of InD points in $\mathcal{S}_{\text{wild}}$ are*
706 *i.i.d., and each coordinate is sub-Gaussian with variance proxy σ^2 . Let $\varepsilon = \sigma \sqrt{2 \log(2dm_{\text{in}})}$, and*
707 *fix any confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification*
708 *rate of the element-wise median (EWM) filtering rule satisfies:*

$$\text{ERR}_{\text{in}} \leq \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}.$$

709 *Proof.* We structure our argument into two key steps. In **Step 1**, we establish a high-probability
710 concentration bound for the gradients of InD samples around their mean, denoted as $\bar{\mathbf{v}}_{\text{in}}$. We define
711 an InD sample whose gradient lies within this high-probability region as a *good inlier*. In **Step**
712 **2**, we leverage a “swapping” argument on the EWM objective to demonstrate that removing more
713 than a $(2\eta + \frac{\pi}{2}(1 - \pi))$ fraction of *good inliers*—where η represents the fraction of InD samples
714 with gradients outside the high-probability region (i.e., non-good InD samples)—would contradict



(a) InD gradient histogram.



(b) InD gradient Q-Q plot.

Figure 4: Illustration of InD sample gradients exhibiting sub-Gaussian behavior in each coordinate. (left) Histogram of gradient values (CIFAR-100 InD data) showing concentration around the mean with light tails, consistent with sub-Gaussianity. (right) Q-Q plot comparing empirical quantiles of InD gradients against a theoretical Gaussian distribution, confirming alignment with sub-Gaussianity.

optimality. Combining these two steps, we derive the stated upper bound on the inlier misclassification rate, ERR_{in} .

Step 1: Concentration of InD Gradients. Let $\nabla \ell(f_\phi(\mathbf{x})) \in \mathbb{R}^d$ be the gradient (with respect to ϕ) of a loss evaluated at an InD sample $\mathbf{x}$. Suppose each coordinate of this gradient is sub-Gaussian with variance proxy σ^2 , centered at the same mean vector $\bar{\nabla}_{\text{in}}$. That is, for each coordinate $j \in \{1, \dots, d\}$,

$$\mathbb{P}\left(|\nabla \ell(f_\phi(\mathbf{x}))_j - \bar{\nabla}_{\text{in},j}| \geq \epsilon\right) \leq 2 \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (6)$$

Define

$$\mathbf{v} := \nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}} \in \mathbb{R}^d.$$

It is straightforward to show that

$$\|\mathbf{v}\|_2 \leq \sqrt{d} \|\mathbf{v}\|_\infty,$$

so if $\|\mathbf{v}\|_2 > \epsilon \sqrt{d}$ then there must exist at least one coordinate j with $|v_j| > \epsilon$.

Hence, combining (6) across $j = 1, \dots, d$ via a union bound yields

$$\mathbb{P}\left(\|\nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}\right) \leq 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (7)$$

Let $m_{\text{in}} = (1 - \pi)m$ denote the total number of InD points in the set $\mathcal{S}_{\text{wild}}$. We define an InD point $\mathbf{x}$ to be *bad* if

$$\|\nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}. \quad (8)$$

The objective is to show that the number of such bad inliers is a small fraction of m_{in} with high probability.

Using the sub-Gaussian assumption together with a union bound over the d coordinates, define $p \in [0, 1]$ by

$$p := \mathbb{P}\left(\|\nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}\right) \leq 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (9)$$

Here, $\mathbf{x}$ denotes a generic InD sample, and the m_{in} inlier samples are assumed to be i.i.d.

Let X denote the number of bad inliers among the m_{in} samples. For each inlier $\mathbf{x}_i$, define the indicator

$$I_i := \begin{cases} 1, & \text{if } \|\nabla \ell(f_\phi(\mathbf{x}_i)) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}, \\ 0, & \text{otherwise.} \end{cases} \quad (10)$$

733 Then X is represented as

$$X = \sum_{i=1}^{m_{\text{in}}} I_i. \quad (11)$$

734 Linearity of expectation gives

$$\mathbb{E}[X] = \sum_{i=1}^{m_{\text{in}}} \mathbb{E}[I_i] \quad (12)$$

$$= \sum_{i=1}^{m_{\text{in}}} \mathbb{P}(\mathbf{x}_i \text{ is bad}) \quad (13)$$

$$\stackrel{\text{from (9)}}{\leq} m_{\text{in}} \cdot p \quad (14)$$

$$\leq m_{\text{in}} \cdot 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (15)$$

735 Since X is a sum of m_{in} independent Bernoulli variables bounded in $[0, 1]$, Hoeffding's inequality
736 yields, for any $t > 0$:

$$\mathbb{P}(X \geq \mathbb{E}[X] + t) \leq \exp\left(-\frac{2t^2}{m_{\text{in}}}\right). \quad (16)$$

737 Setting

$$t := \sqrt{\frac{m_{\text{in}} \log(\frac{1}{\delta})}{2}}, \quad (17)$$

738 this choice gives

$$\mathbb{P}\left(X \geq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}\right) \leq \delta. \quad (18)$$

739 Therefore, with probability at least $1 - \delta$:

$$X \leq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}. \quad (19)$$

740 Dividing by m_{in} yields the following upper bound on the bad-inlier fraction:

$$\frac{X}{m_{\text{in}}} \leq \frac{\mathbb{E}[X]}{m_{\text{in}}} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \quad (20)$$

$$\stackrel{\text{from (12)}}{\leq} 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \quad (21)$$

$$= \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}. \quad (22)$$

741 Denote this upper bound by η . With $\epsilon = \sigma\sqrt{2\log(2dm_{\text{in}})}$, it follows that, with probability at least
742 $1 - \delta$, only an $O\left(d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right) + \sqrt{\frac{\log(1/\delta)}{m_{\text{in}}}}\right)$ fraction of inliers have gradients outside the radius
743 $\epsilon\sqrt{d}$ around $\bar{\mathbf{V}}_{\text{in}}$. These points are the *bad inliers*; the remaining InD points are *good inliers*.

744 **Step 2: Contradiction via a Swapping Argument** We show that if $\mathcal{S}^*$ excludes too many InD
 745 points, namely if

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (23)$$

746 then one can construct a replacement set $\hat{\mathcal{S}}$ whose EWM is strictly closer to $\bar{\nabla}_{\text{in}}$ than the EWM of
 747 $\mathcal{S}^*$, contradicting optimality of $\mathcal{S}^*$.

748 Let m_ε be the number of indices i such that either x_i is OOD or x_i is an InD point whose gradient
 749 lies outside the closed ℓ_2 ball

$$B_\varepsilon = \left\{ g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \right\}. \quad (24)$$

750 Let m_{in}^* denote the number of InD points contained in the candidate minimizer $\mathcal{S}^*$.

751 Assume for contradiction that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (25)$$

752 Define the swap size as

$$m'_\varepsilon := \min \{m_\varepsilon, (\text{ERR}_{\text{in}} - \eta)m_{\text{in}}\}. \quad (26)$$

753 This quantity is the number of good inliers, i.e., points inside B_ε , that are excluded from $\mathcal{S}^*$ and can
 754 be inserted by swapping.

755 Construct $\hat{\mathcal{S}}$ by removing m'_ε points in $\mathcal{S}^*$ that lie outside B_ε , either OOD points or bad inliers, and
 756 inserting m'_ε good inliers from B_ε . This operation preserves cardinality:

$$|\hat{\mathcal{S}}| = |\mathcal{S}^*|.$$

757 Recall that

$$\text{ERR}_{\text{in}} := \frac{m_\varepsilon - m_{\text{in}}^*}{m_{\text{in}}}, \quad (27)$$

758 which is the fraction of good inliers excluded from $\mathcal{S}^*$. Rearranging gives

$$m_\varepsilon - m_{\text{in}}^* = \text{ERR}_{\text{in}} \cdot m_{\text{in}}. \quad (28)$$

759 Under the contradiction assumption,

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (29)$$

760 Subtracting η and multiplying by m_{in} gives

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}}. \quad (30)$$

761 At most an η fraction of inliers lie outside B_ε , and the OOD fraction is π . Hence the total number of
 762 samples outside B_ε is bounded by

$$m_\varepsilon \leq \eta m_{\text{in}} + \frac{\pi}{1-\pi} m_{\text{in}} = \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}}. \quad (31)$$

763 It follows that

$$\left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}} > \frac{1}{2} \cdot \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}} \geq \frac{m_\varepsilon}{2}. \quad (32)$$

764 Combining the inequalities yields

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \frac{m_\varepsilon}{2}, \quad (33)$$

765 Thus $m'_\varepsilon = m_\varepsilon/2 + \eta m_{\text{in}}$, implying

$$m_\varepsilon - m'_\varepsilon = \frac{m_\varepsilon}{2} - \eta m_{\text{in}}, \quad (34)$$

$$m_{\text{in}}^* + m'_\varepsilon = m_{\text{in}}^* + \frac{m_\varepsilon}{2} + \eta m_{\text{in}}. \quad (35)$$

766 Because $\eta m_{\text{in}} > 0$, it follows that

$$m_{\text{in}}^* + m'_\varepsilon > m_\varepsilon - m'_\varepsilon, \quad (36)$$

767 so points inside B_ε constitute a strict majority of $\widehat{\mathcal{S}}$.

768 Fix any coordinate $j \in \{1, \dots, d\}$. Let $X^{(j)} = \{x_1^{(j)}, \dots, x_m^{(j)}\}$ denote the j -th coordinate values
769 of gradients in $\mathcal{S}^*$, and let $\widehat{X}^{(j)} = \{\widehat{x}_1^{(j)}, \dots, \widehat{x}_m^{(j)}\}$ denote the corresponding values in $\widehat{\mathcal{S}}$. Let
770 $\mu_j = \bar{\nabla}_{\text{in},j}$.

771 Each inserted gradient lies inside the interval $I_j = [\mu_j - \varepsilon, \mu_j + \varepsilon]$, and each removed gradient lies
772 outside I_j . For at least one coordinate j^* , the swaps increase the number of values in I_{j^*} from below
773 $m/2$ to above $m/2$. Therefore,

$$\text{med}(X^{(j^*)}) \notin I_{j^*}, \quad \text{med}(\widehat{X}^{(j^*)}) \in I_{j^*}, \quad (37)$$

774 which implies

$$|\text{med}(\widehat{X}^{(j^*)}) - \mu_{j^*}| \leq \varepsilon \quad \text{and} \quad |\text{med}(\widehat{X}^{(j^*)}) - \mu_{j^*}| < |\text{med}(X^{(j^*)}) - \mu_{j^*}|. \quad (38)$$

775 For all remaining coordinates $j \neq j^*$, the swaps either preserve or increase the in-interval majority,
776 so the median distance from μ_j cannot increase.

777 Consequently, coordinate-wise:

$$|\text{EWM}_j(G_{\widehat{\mathcal{S}}}) - \mu_j| \leq |\text{EWM}_j(G_{\mathcal{S}^*}) - \mu_j|, \quad (39)$$

778 with strict inequality for at least one coordinate j^* , which implies

$$\|\text{EWM}(G_{\widehat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2 < \|\text{EWM}(G_{\mathcal{S}^*}) - \bar{\nabla}_{\text{in}}\|_2. \quad (40)$$

779 This contradicts the assumed optimality of $\mathcal{S}^*$ and completes the proof.

$$\Rightarrow \quad \text{ERR}_{\text{in}} \leq 2\eta + \frac{\pi}{2(1-\pi)}. \quad (41)$$

780 □

781 C.2 Bound on the Outlier Retention Rate

782 We now derive a non-asymptotic upper bound on the fraction of OOD points retained by the MEDIX
783 filter. The proof parallels the inlier misclassification analysis (Appendix C.1), but replaces InD
784 concentration around the reference gradient with a separation condition between OOD and InD
785 gradient means.

786 **Theorem C.2** (Outlier Retention Bound under Vector Separation). Assume that for every OOD point
 787 $x \sim P_{\text{out}}$, the gradient $g(x) := \nabla \ell(f_{\varphi_{\text{in}}}(x)) \in \mathbb{R}^d$ satisfies:

788 **(I) Sub-Gaussianity.** Each coordinate $g_j(x)$ is sub-Gaussian with variance proxy σ_{out}^2 .

789 **(II) Vector Mean Separation.** The mean OOD gradient vector μ_{out} satisfies

$$\|\mu_{\text{out}} - \bar{\nabla}_{\text{in}}\|_2 \geq \Delta\sqrt{d}, \quad \text{for some } \Delta > 0.$$

790 Fix any tolerance $\varepsilon \in (0, \Delta)$ and confidence parameter $\delta \in (0, 1)$. Define:

$$p_{\text{out}}(\varepsilon) := 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right), \quad \eta_{\text{out}} := p_{\text{out}}(\varepsilon) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}.$$

791 Then with probability at least $1 - \delta$, the outlier retention rate satisfies:

$$\text{ERR}_{\text{out}} := \frac{|\mathcal{S}_{\text{in}}^* \cap \text{OOD}|}{m_{\text{out}}} \leq 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}} + \frac{1 - \pi}{2\pi}. \quad (42)$$

792 *Proof. Step 1: Concentration of OOD Gradients.*

793 By assumption, each coordinate of $g(x)$ is sub-Gaussian and the OOD mean vector is separated from
 794 $\bar{\nabla}_{\text{in}}$ by at least $\Delta\sqrt{d}$. Therefore, for any OOD point x , the event

$$\|g(x) - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \quad (43)$$

795 has probability no greater than $p_{\text{out}}(\varepsilon)$.

796 **Counting the “ambiguous” OOD points.** For every OOD example x_i define the indicator

$$J_i := \mathbf{1}\left\{\|g(x_i) - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d}\right\}, \quad i = 1, \dots, m_{\text{out}}.$$

797 Each $J_i \sim \text{Bernoulli}(p_{\text{out}}(\varepsilon))$ and the J_i ’s are independent. Let

$$Y := \sum_{i=1}^{m_{\text{out}}} J_i, \quad \mu := \mathbb{E}[Y] = m_{\text{out}} p_{\text{out}}(\varepsilon).$$

798 Because the indicators J_i are bounded in $\{0, 1\}$, Hoeffding’s inequality gives, for any $s > 0$,

$$\Pr\{Y - \mu \geq s\} \leq \exp\left(-\frac{2s^2}{m_{\text{out}}}\right).$$

799 Choose $s = \sqrt{\frac{m_{\text{out}} \log(1/\delta)}{2}}$; then (C.2) equals δ , and with probability at least $1 - \delta$

$$Y \leq m_{\text{out}} \left(p_{\text{out}}(\varepsilon) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}\right) = m_{\text{out}} \eta_{\text{out}}.$$

800 Define the high-probability event

$$\mathcal{E}_1 := \{Y \leq m_{\text{out}} \eta_{\text{out}}\}, \quad (35)$$

801 which holds with probability at least $1 - \delta$.

802 **Step 2: Contradiction via a Swapping Argument** Suppose, toward contradiction, that

$$\text{ERR}_{\text{out}} > \eta_{\text{out}} + \frac{1 - \pi}{2\pi}. \quad (44)$$

803 Let $m_{\text{out}}^* := |\mathcal{S}_{\text{in}}^* \cap \text{OOD}|$, and define $m_{\varepsilon}^* := \#\{\text{ambiguous OODs in } \mathcal{S}_{\text{in}}^*\}$. Then, under $\mathcal{E}_1$,

$$m_{\text{sep}}^* := m_{\text{out}}^* - m_\varepsilon^* > \frac{1-\pi}{2\pi} m_{\text{out}} = \frac{1}{2} m_{\text{in}}. \quad (45)$$

Thus, more than half of the points in $\mathcal{S}_{\text{in}}^*$ are separated OOD samples with gradients outside the ball

$$B_\varepsilon := \left\{ g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \right\}. \quad (46)$$

By the definition of the coordinate-wise median, the majority condition implies that, in every coordinate, most entries of $G_{\mathcal{S}_{\text{in}}^*}^{(j)}$ lie outside $[\bar{\nabla}_{\text{in},j} - \varepsilon, \bar{\nabla}_{\text{in},j} + \varepsilon]$. Hence,

$$\|\text{EWM}(G_{\mathcal{S}_{\text{in}}^*}) - \bar{\nabla}_{\text{in}}\|_2^2 \geq \varepsilon^2 d. \quad (47)$$

Now construct a subset $\hat{\mathcal{S}}$ by replacing the m_{sep}^* separated OOD samples in $\mathcal{S}_{\text{in}}^*$ with the same number of InD samples whose gradients lie in B_ε . Such inliers exist because fewer than $m_{\text{out}}\eta_{\text{out}}$ OOD points are ambiguous and the InD pool contains m_{in} samples.

After this swap, at least half of the entries in each coordinate of $G_{\hat{\mathcal{S}}}^{(j)}$ lie in $[\bar{\nabla}_{\text{in},j} - \varepsilon, \bar{\nabla}_{\text{in},j} + \varepsilon]$, which implies

$$\|\text{EWM}(G_{\hat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2^2 \leq \varepsilon^2 d. \quad (48)$$

Combining this inequality with (47) gives the strict improvement

$$\|\bar{\nabla}_{\text{in}} - \text{EWM}(G_{\hat{\mathcal{S}}})\|_2^2 < \|\bar{\nabla}_{\text{in}} - \text{EWM}(G_{\mathcal{S}_{\text{in}}^*})\|_2^2. \quad (49)$$

This contradicts the optimality of $\mathcal{S}_{\text{in}}^*$. Hence the contradiction assumption is false, and with probability at least $1 - \delta$, the outlier retention rate is bounded by

$$\text{ERR}_{\text{out}} \leq 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}} + \frac{1-\pi}{2\pi}, \quad (50)$$

□

C.3 Theorem C.3 without Sub-Gaussianity

Theorem C.3 (Inlier Misclassification Bound without Sub-Gaussianity). *Assume that the gradients of InD points in $\mathcal{S}_{\text{wild}}$ are i.i.d., and that each coordinate has variance σ^2 and finite fourth moment bounded by μ_4 , i.e.,*

$$\mathbb{E}[(\nabla\ell(f_\phi(x)))_j - \bar{\nabla}_{\text{in},j}]^2 = \sigma^2, \quad \mathbb{E}[(\nabla\ell(f_\phi(x)))_j - \bar{\nabla}_{\text{in},j}]^4 \leq \mu_4.$$

Let $m_{\text{in}} = (1 - \pi)m$ denote the number of InD samples in $\mathcal{S}_{\text{wild}}$, and fix any tolerance $\varepsilon > \sigma$ and confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification rate of the element-wise median (EWM) filtering rule satisfies:

$$\text{ERR}_{\text{in}} \leq \underbrace{2 \left(\frac{\mu_4 - \sigma^4}{d(\varepsilon^2 - \sigma^2)^2} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \right)}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}.$$

823 **Differences with the original bound.** The sub-Gaussian bound contains the decay term

$$2 \exp \left(-\frac{\epsilon^2 d}{2\sigma_{\text{sub}}^2} \right)$$

824 which decreases exponentially with dimension d . The fourth-moment bound instead contains the
825 term

$$\frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}$$

826 which decays as $O(1/d)$ and is therefore looser in high dimension. The original result requires
827 sub-Gaussian tails, whereas the modified result only assumes finite fourth moments and thus permits
828 heavier-tailed gradient coordinates. Both bounds retain the finite-sample term

$$\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}},$$

829 which decreases with m_{in} independently of the tail assumption.

830 *Proof.* The proof follows the structure of the sub-Gaussian theorem. First, it establishes a concen-
831 tration bound on the fraction of “bad” InD samples, namely samples whose gradients are far from
832 ∇_{in} , without invoking sub-Gaussian tails. Second, it applies the same swapping argument to control
833 the inlier misclassification rate. The only substantive change is that the sub-Gaussian tail estimate is
834 replaced by a moment-based bound derived from the finite fourth-moment assumption.

835 **Step 1: Concentration of InD Gradients** Define the gradient deviation for an InD sample x as:

$$v = \nabla \ell(f_\phi(x)) - \nabla_{\text{in}} \in \mathbb{R}^d, \quad (51)$$

836 where $\nabla_{\text{in}} = \frac{1}{n} \sum_{(x_i, y_i) \in S_{\text{in}}} \nabla \ell(f_\phi(x_i))$ is the mean gradient over the labeled InD training set S_{in} ,
837 and we assume ∇_{in} is the population mean. An InD sample is classified as “bad” if:

$$\|v\|_2 > \epsilon \sqrt{d}, \quad (52)$$

838 and “good” otherwise. Our goal is to bound the fraction of bad inliers among the m_{in} InD samples in
839 S_{wild} .

840 Since $\|v\|_2^2 = \sum_{j=1}^d v_j^2$, we analyze the ℓ_2 norm via the sum of squared coordinates. Assume that
841 the coordinates $v_j = \nabla \ell(f_\phi(x))_j - \nabla_{\text{in},j}$ are independent across $j = 1, \dots, d$, each with:

$$\mathbb{E}[v_j] = 0 \quad (\text{assuming } \nabla_{\text{in},j} = \mathbb{E}[\nabla \ell(f_\phi(x))_j]), \quad (53)$$

$$\mathbb{E}[v_j^2] = \sigma^2, \quad \mathbb{E}[v_j^4] \leq \mu_4. \quad (54)$$

842 Thus:

$$\mathbb{E}[\|v\|_2^2] = \mathbb{E} \left[\sum_{j=1}^d v_j^2 \right] = d\sigma^2, \quad (55)$$

$$\text{Var}(\|v\|_2^2) = \text{Var} \left(\sum_{j=1}^d v_j^2 \right) = \sum_{j=1}^d \text{Var}(v_j^2). \quad (56)$$

843 Compute the variance of v_j^2 :

$$\text{Var}(v_j^2) = \mathbb{E}[v_j^4] - (\mathbb{E}[v_j^2])^2 \leq \mu_4 - \sigma^4, \quad (57)$$

844 so:

$$\text{Var}(\|v\|_2^2) \leq d(\mu_4 - \sigma^4). \quad (58)$$

845 Using Chebyshev's inequality for $\|v\|_2^2$:

$$\mathbb{P}(|\|v\|_2^2 - d\sigma^2| \geq t) \leq \frac{d(\mu_4 - \sigma^4)}{t^2}. \quad (59)$$

846 We need $\mathbb{P}(\|v\|_2 > \epsilon\sqrt{d})$, which corresponds to $\|v\|_2^2 > \epsilon^2 d$. Set the threshold:

$$\|v\|_2^2 > \epsilon^2 d \implies \|v\|_2^2 - d\sigma^2 > \epsilon^2 d - d\sigma^2 = d(\epsilon^2 - \sigma^2). \quad (60)$$

847 Thus:

$$\mathbb{P}(\|v\|_2^2 > \epsilon^2 d) = \mathbb{P}(\|v\|_2^2 - d\sigma^2 > d(\epsilon^2 - \sigma^2)) \leq \frac{d(\mu_4 - \sigma^4)}{[d(\epsilon^2 - \sigma^2)]^2} = \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}, \quad (61)$$

848 provided $\epsilon > \sigma$ so that $\epsilon^2 - \sigma^2 > 0$. Define:

$$p = \mathbb{P}(\|\nabla \ell(f_\phi(x)) - \nabla_{\text{in}}\|_2 > \epsilon\sqrt{d}) \leq \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}. \quad (62)$$

849 Let X be the number of bad inliers among the m_{in} InD samples in S_{wild} , where each InD sample is
850 bad with probability at most p . Then:

$$\mathbb{E}[X] \leq m_{\text{in}}p \leq m_{\text{in}} \cdot \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}. \quad (63)$$

851 Since $X = \sum_{i=1}^{m_{\text{in}}} I_i$, where $I_i = 1$ if the i -th InD sample is bad and 0 otherwise, and I_i are i.i.d.
852 Bernoulli(p) variables bounded in $[0, 1]$, apply Hoeffding's inequality:

$$\mathbb{P}(X \geq \mathbb{E}[X] + t) \leq \exp\left(-\frac{2t^2}{m_{\text{in}}}\right). \quad (64)$$

853 Set $t = \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}$, so:

$$\mathbb{P}\left(X \geq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}\right) \leq \delta. \quad (65)$$

854 With probability at least $1 - \delta$:

$$X \leq m_{\text{in}} \cdot \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2} + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}. \quad (66)$$

855 The fraction of bad inliers is:

$$\eta = \frac{X}{m_{\text{in}}} \leq \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}. \quad (67)$$

856 This η represents the upper bound on the fraction of InD samples whose gradients deviate from $\bar{\nabla}_{\text{in}}$
857 by more than $\epsilon\sqrt{d}$, relying only on finite fourth moments and coordinate independence.

858 **Step 2: Contradiction via a Swapping Argument** We show that if $\mathcal{S}^*$ excludes too many InD
859 points, namely if

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (68)$$

860 then one can construct a replacement set $\hat{\mathcal{S}}$ whose EWM is strictly closer to $\bar{\nabla}_{\text{in}}$ than the EWM of
861 $\mathcal{S}^*$, contradicting optimality of $\mathcal{S}^*$.

862 Let m_ϵ be the number of indices i such that either x_i is OOD or x_i is an InD point whose gradient
863 lies outside the closed ℓ_2 ball

$$B_\epsilon = \left\{ g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \epsilon\sqrt{d} \right\}. \quad (69)$$

864 Let m_{in}^* denote the number of InD points contained in the candidate minimizer $\mathcal{S}^*$.

865 Assume for contradiction that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (70)$$

866 Define the swap size as

$$m'_\epsilon := \min \{ m_\epsilon, (\text{ERR}_{\text{in}} - \eta)m_{\text{in}} \}. \quad (71)$$

867 This quantity is the number of good inliers, i.e., points inside B_ϵ , that are excluded from $\mathcal{S}^*$ and can
868 be inserted by swapping.

869 Construct $\hat{\mathcal{S}}$ by removing m'_ϵ points in $\mathcal{S}^*$ that lie outside B_ϵ , either OOD points or bad inliers, and
870 inserting m'_ϵ good inliers from B_ϵ . This operation preserves cardinality:

$$|\hat{\mathcal{S}}| = |\mathcal{S}^*|.$$

871 Recall that

$$\text{ERR}_{\text{in}} := \frac{m_\epsilon - m_{\text{in}}^*}{m_{\text{in}}}, \quad (72)$$

872 which is the fraction of good inliers excluded from $\mathcal{S}^*$. Rearranging gives

$$m_\epsilon - m_{\text{in}}^* = \text{ERR}_{\text{in}} \cdot m_{\text{in}}. \quad (73)$$

873 Under the contradiction assumption,

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (74)$$

874 Subtracting η and multiplying by m_{in} gives

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}}. \quad (75)$$

At most an η fraction of inliers lie outside B_ε , and the OOD fraction is π . Hence the total number of samples outside B_ε is bounded by

$$m_\varepsilon \leq \eta m_{\text{in}} + \frac{\pi}{1-\pi} m_{\text{in}} = \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}}. \quad (76)$$

It follows that

$$\left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}} > \frac{1}{2} \cdot \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}} \geq \frac{m_\varepsilon}{2}. \quad (77)$$

Combining the inequalities yields

$$(\text{ERR}_{\text{in}} - \eta) m_{\text{in}} > \frac{m_\varepsilon}{2}, \quad (78)$$

Thus $m'_\varepsilon = m_\varepsilon/2 + \eta m_{\text{in}}$, implying

$$m_\varepsilon - m'_\varepsilon = \frac{m_\varepsilon}{2} - \eta m_{\text{in}}, \quad (79)$$

$$m_{\text{in}}^* + m'_\varepsilon = m_{\text{in}}^* + \frac{m_\varepsilon}{2} + \eta m_{\text{in}}. \quad (80)$$

Because $\eta m_{\text{in}} > 0$, it follows that

$$m_{\text{in}}^* + m'_\varepsilon > m_\varepsilon - m'_\varepsilon, \quad (81)$$

so points inside B_ε constitute a strict majority of $\widehat{\mathcal{S}}$.

Fix any coordinate $j \in \{1, \dots, d\}$. Let $X^{(j)} = \{x_1^{(j)}, \dots, x_m^{(j)}\}$ denote the j -th coordinate values of gradients in $\mathcal{S}^*$, and let $\widehat{X}^{(j)} = \{\widehat{x}_1^{(j)}, \dots, \widehat{x}_m^{(j)}\}$ denote the corresponding values in $\widehat{\mathcal{S}}$. Let $\mu_j = \bar{\nabla}_{\text{in},j}$.

Each inserted gradient lies inside the interval $I_j = [\mu_j - \varepsilon, \mu_j + \varepsilon]$, and each removed gradient lies outside I_j . For at least one coordinate j^* , the swaps increase the number of values in I_{j^*} from below $m/2$ to above $m/2$. Therefore,

$$\text{med}(X^{(j^*)}) \notin I_{j^*}, \quad \text{med}(\widehat{X}^{(j^*)}) \in I_{j^*}, \quad (82)$$

which implies

$$|\text{med}(\widehat{X}^{(j^*)}) - \mu_{j^*}| \leq \varepsilon \quad \text{and} \quad |\text{med}(\widehat{X}^{(j^*)}) - \mu_{j^*}| < |\text{med}(X^{(j^*)}) - \mu_{j^*}|. \quad (83)$$

For all remaining coordinates $j \neq j^*$, the swaps either preserve or increase the in-interval majority, so the median distance from μ_j cannot increase.

Consequently, coordinate-wise:

$$|\text{EWM}_j(G_{\widehat{\mathcal{S}}}) - \mu_j| \leq |\text{EWM}_j(G_{\mathcal{S}^*}) - \mu_j|, \quad (84)$$

with strict inequality for at least one coordinate j^* , which implies

$$\|\text{EWM}(G_{\widehat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2 < \|\text{EWM}(G_{\mathcal{S}^*}) - \bar{\nabla}_{\text{in}}\|_2. \quad (85)$$

This contradicts the assumed optimality of $\mathcal{S}^*$ and completes the proof.

$$\Rightarrow \quad \text{ERR}_{\text{in}} \leq 2\eta + \frac{\pi}{2(1-\pi)}. \quad (86)$$

□